Refuelling your Jetpack

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What is Android Jetpack?

- Suite of **libraries**, **tools**, and **guidance** for less boilerplate and a clearer structure
- New features without waiting for users to upgrade their devices to new Android versions
- Instead of ***Do it however you want***, Google has an **opinion** about how things should be done
- Backward compatibility built in (fragmentation and OS differences are handled for you)

The Era of Fragmentation

- New features used to be tied to OS updates, but OS updates took ages to reach devices
- The Android Support Library (`android.support.*`) was Google's band-aid. It sort of helped, but…
- …it turned into a mess. `v4`, `v7`, `v13`… jar files everywhere
- A dependency hell

The Reboot – Enter Android Jetpack

- 2017: Architecture Components (*ViewModel*, *Room*, *Lifecycle*, *LiveData*) announced at I/O; 1.0 that November
- 2018: Jetpack and the great reboot to `androidx.*`
  - libraries unbundled from the OS
  - Own release cycles
  - Semantic versioning

Google got opinionated about architecture

From
Do it however it works for you
to
Here's what we think works best

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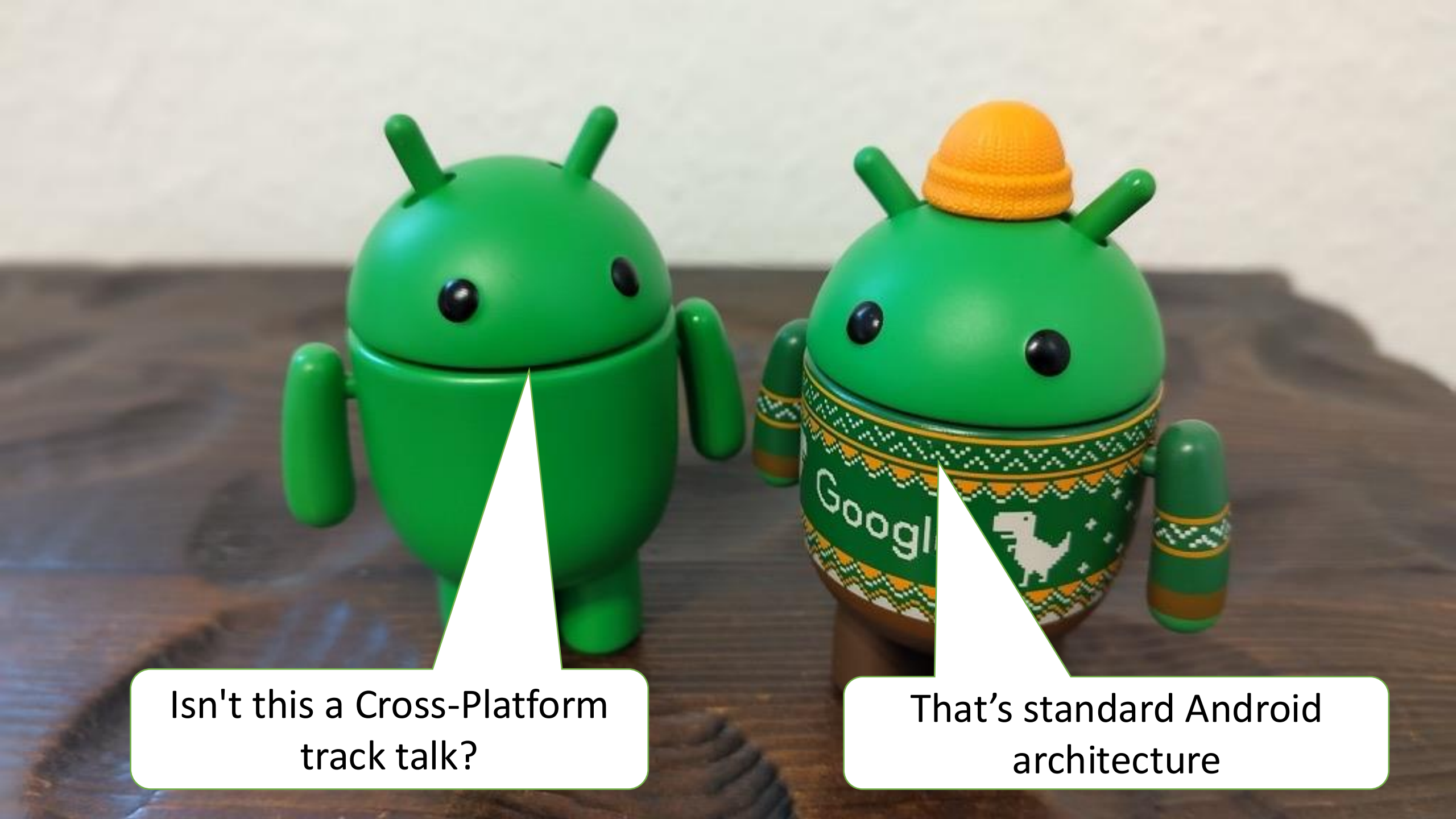
Fast forward to the Jetpack Jungle

- Today there are 130+ artifacts
- Not just modularity; instead, it's baggage
- Zombie libraries: deprecated, duplicate, still haunting the docs e.g. `lifecycle-extensions` (deprecated, split into separate `lifecycle-*` libs), `security-crypto` (deprecated, no clear successor)
- How to build a modern architecture with three ways to do the same thing?

The Official Architecture

- Google's answer to the chaos: a curated stack (the Golden Path)
- UI reacts to state, events flow up. ViewModels hold state and survive config changes
- One navigation graph. One back stack. No reimplementing per screen
- Reactive, typed persistence: Room for the DB, DataStore for preferences / configuration

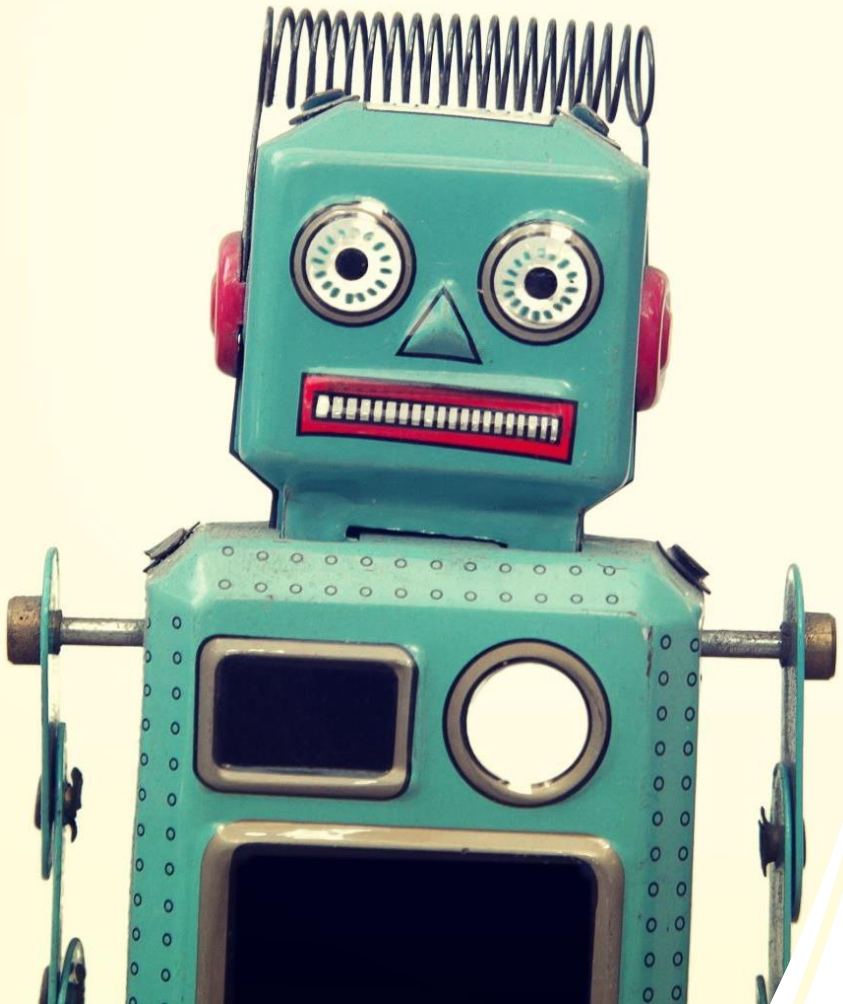

Isn't this a Cross-Platform track talk?
That’s standard Android architecture

The Twist

- That stack you just saw runs on iOS, Desktop, and Web
- Same APIs
- One architecture, the same code

Compose Multiplatform

- Same Jetpack Compose that every Android dev loves, just unchained from Android
- Declarative, reactive: one mental model for every screen
- On iOS and Desktop, it uses Skia and runs natively
- No more learning three UI frameworks
- Android skills deploy to iOS, Desktop, and Web



State & Lifecycle: (finally) adopting the Standard

- Android didn't invent the ViewModel; others had it years before
- An idea too good to neglect
- You put ViewModels in `commonMain`
- They survive config changes (e. g. rotation) and backgrounding on all platforms

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Navigation

- With Navigation Compose you typesafe-ly define your graph and routes once in `commonMain`
- Back stack and deep links work the same on Android, iOS, and Desktop
- Activities, View Controllers, and windows are just thin hosts
- Instead, shared code drives the UI

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Persistence: Room

- *Room* is an ORM on SQLite: Kotlin classes are mapped to tables, DAOs, Flow for reactive queries
- Compile-time SQL verification: no nasty runtime surprises. Runs on each platform's native SQLite
- One database layer in shared code
- ViewModels or repositories consume it the same way everywhere

Configuration & Preferences: DataStore

- *DataStore* allows typed, async reading / writing of preferences and configuration files
- Not blocking the UI thread
- Choose between key-value or protocol buffers. Same API on every platform. Settings, feature flags, preferences—all in shared code
- No more `UserDefaults` for iOS and `SharedPreferences` for Android

One architecture, (at least) three platforms

- The shared module holds almost everything: UI, ViewModels, navigation, Room, DataStore
- Platform apps are thin shells: they just bootstrap

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The *shared* module structure

- `commonMain` contains Compose UI, ViewModels, Navigation, Room, DataStore
- When you need something platform-specific (file paths, open URL, back button), you use `expect` (interface) / `actual` (implementation)

Platform Entry Points

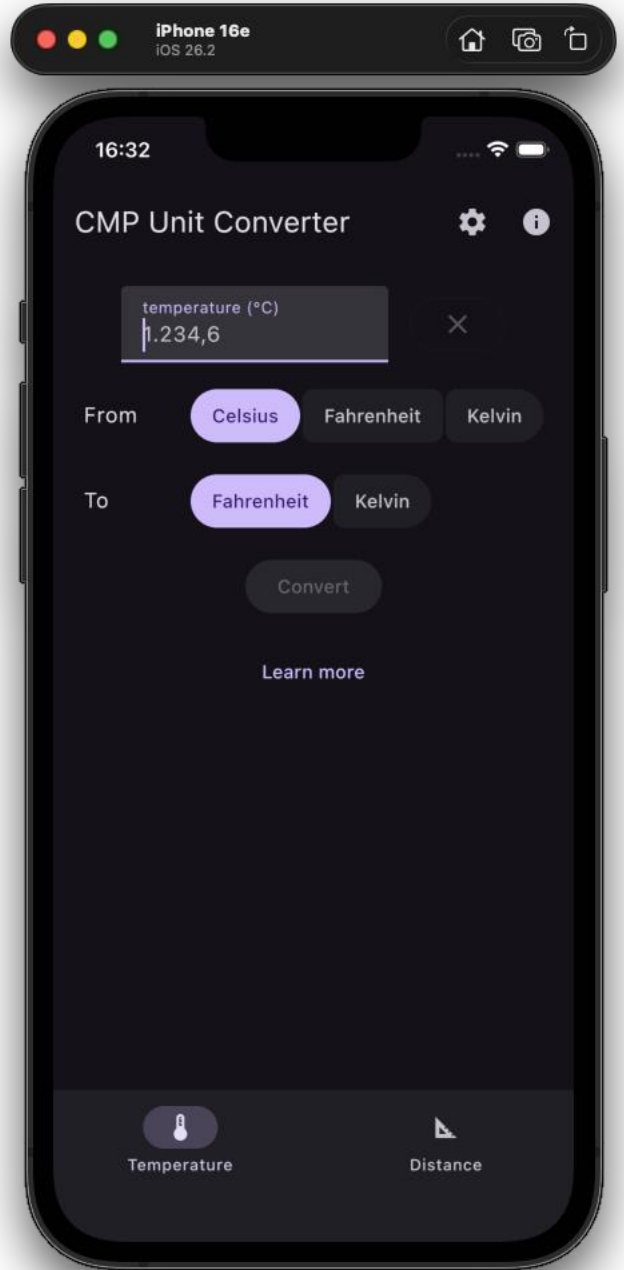
- Android: `Activity` and `setContent`
- iOS: a view controller hosting Compose
- Desktop: `main()` and one or more windows
- Each one wires lifecycle and hosts the same root composable
- You initialize your dependency injection framework at startup; the rest is shared

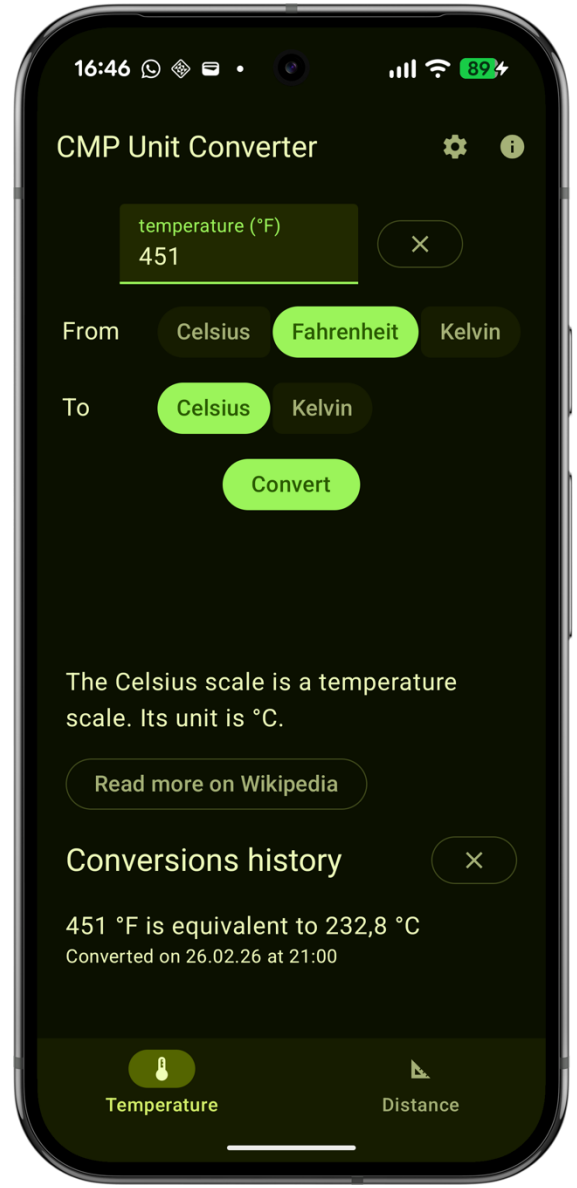
Dependency Injection: The Hilt Question

- Hilt is the recommended DI on Android. But Hilt is Dagger and annotation processing
- It doesn't run on iOS or Desktop
- Google hasn't announced a multiplatform roadmap for Hilt. For shared code today, we use a pure Kotlin option (for example, Koin)
- You define modules in shared code; platform supplies the actuals (e.g. where's the data directory). One container, initialized once per platform

CMP Unit Converter

- A real app: unit conversion (temperature, distance, etc.) on Android, iOS, and Desktop
- Built on the full *golden path*: Compose, ViewModel, Room, DataStore, Koin
- One codebase, minimal platform glue





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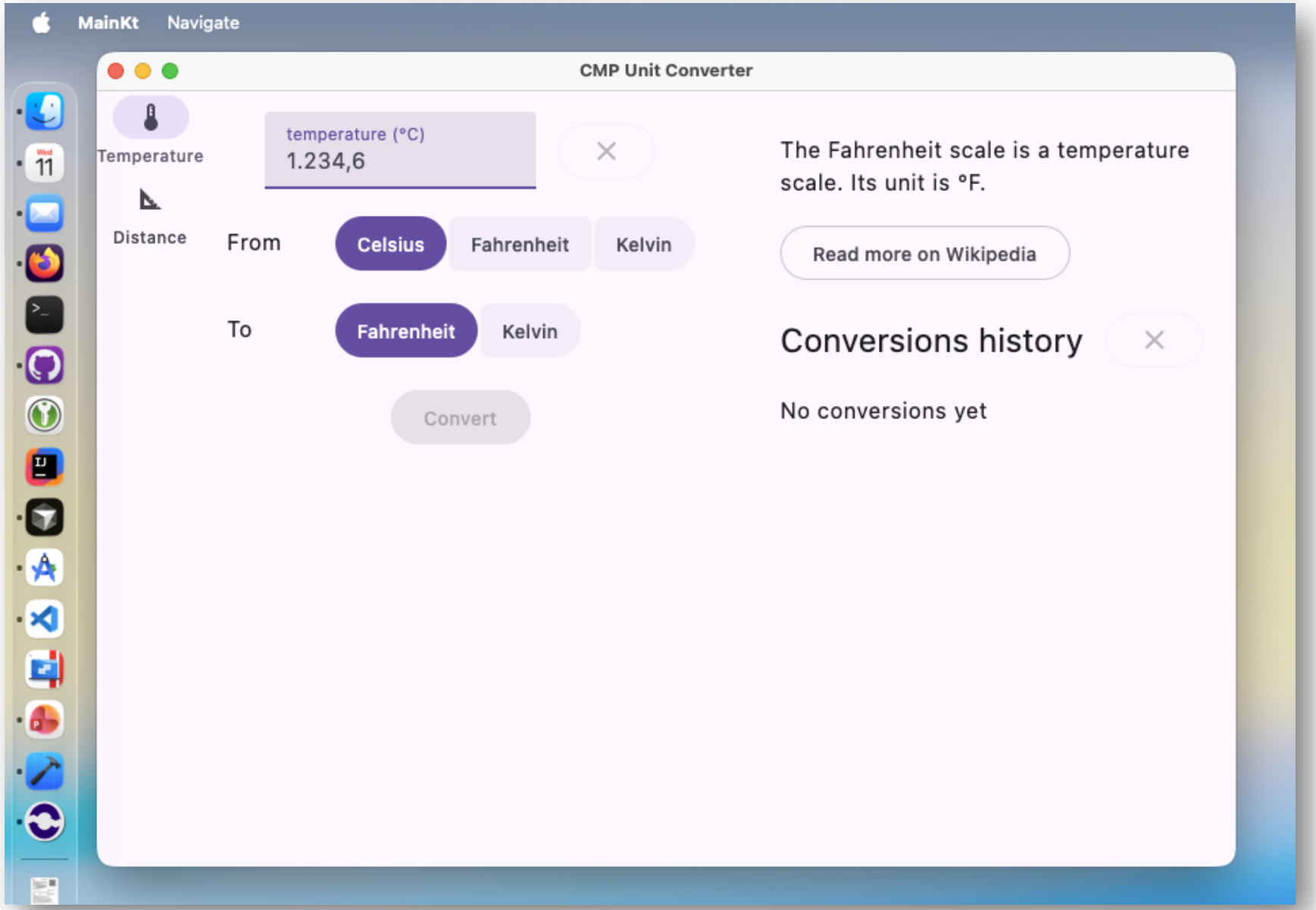

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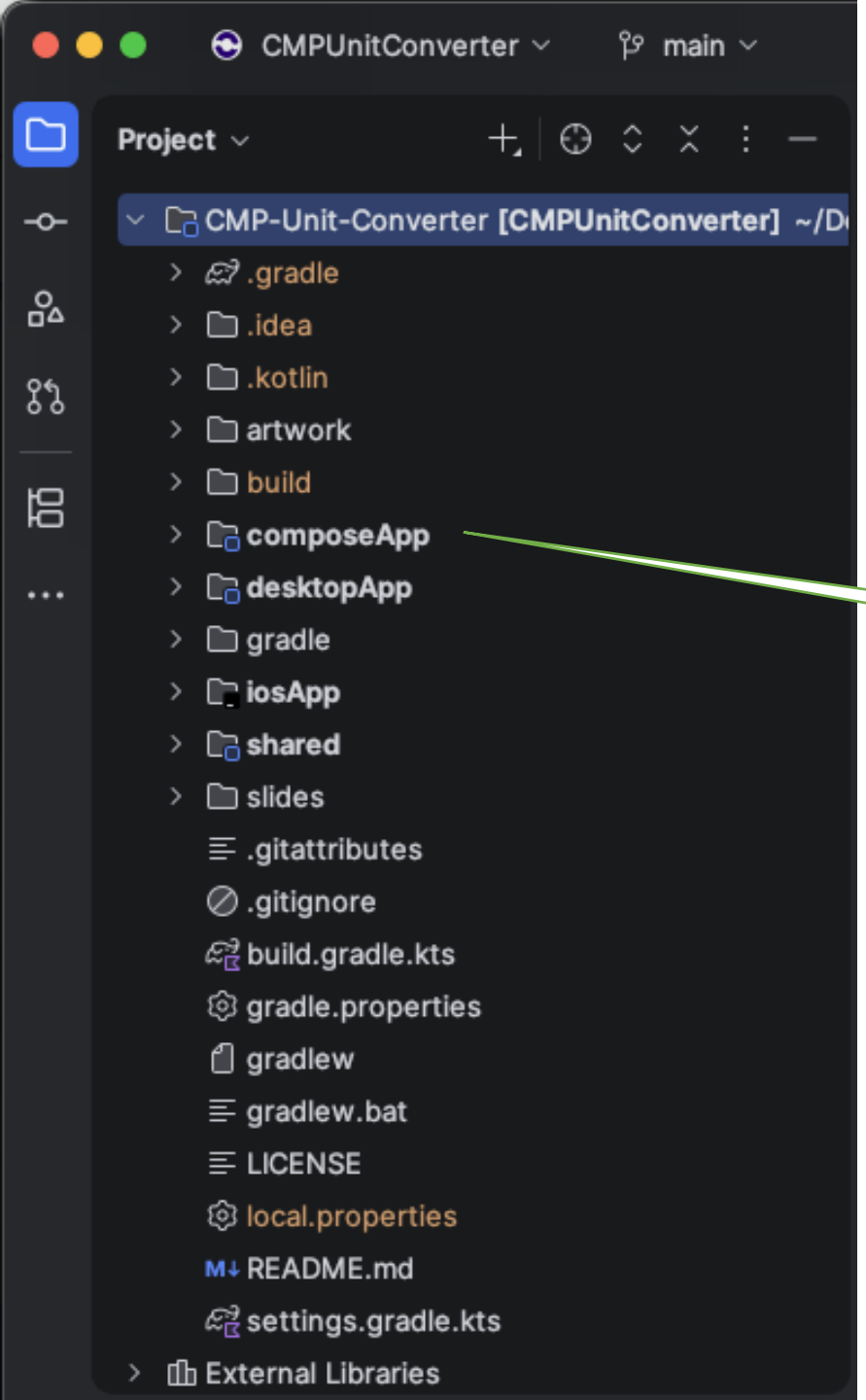
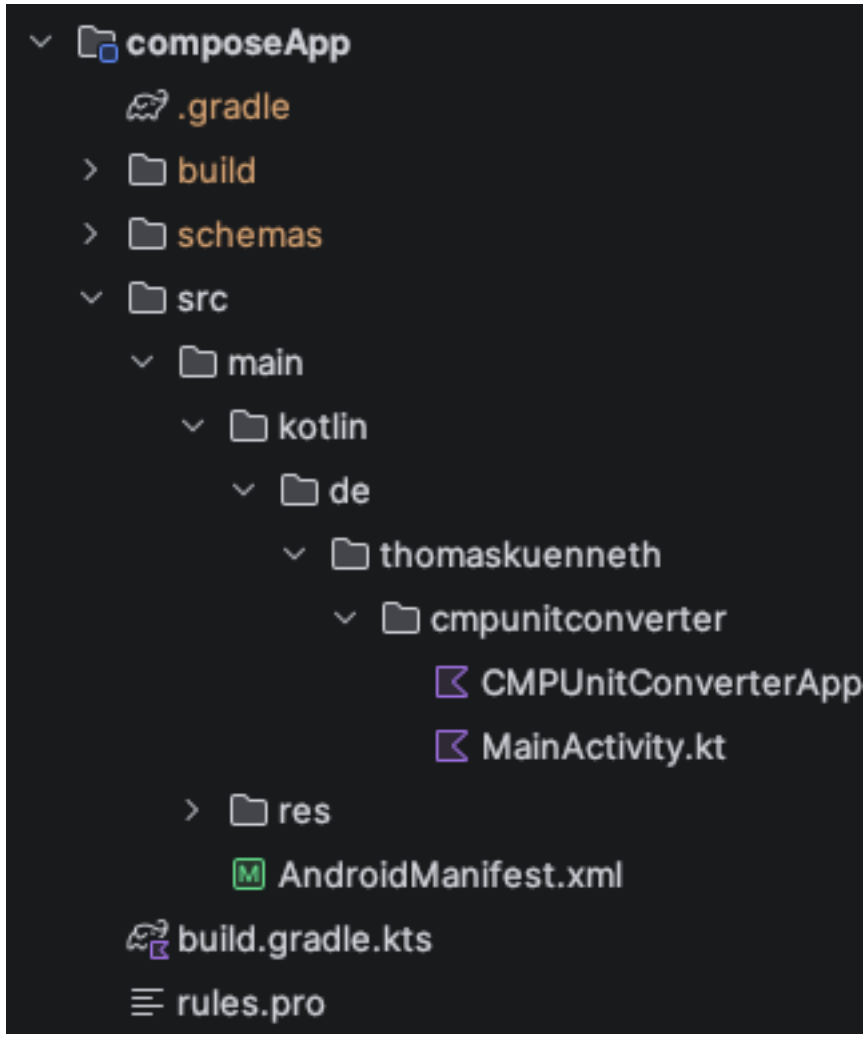
https://github.com/tkuenneth/CMP-Unit-Converter



Android



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CMPUnitConverter main

Project

- CMP-Unit-Converter [CMPUnitConverter] ~/D
  - .gradle
  - .idea
  - .kotlin
  - artwork
  - build
  - composeApp
  - desktopApp
  - gradle
  - iosApp
  - shared
  - slides
  - .gitattributes
  - .gitignore
  - build.gradle.kts
  - gradle.properties
  - gradlew
  - gradlew.bat
  - LICENSE
  - local.properties
  - README.md
  - settings.gradle.kts
- External Libraries

Desktop

- desktopApp
  - build
  - src
    - main
      - kotlin
        - de.thomaskuenneth.cmpunitcor
          - AboutWindow.kt
          - main.kt
          - SettingsWindow.kt
  - build.gradle.kts
  - rules.pro

Experts

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  - gradlew.bat
  - LICENSE
  - local.properties
  - README.md
  - settings.gradle.kts
- External Libraries

iOS

- iosApp
  - iosApp ~/Development/Personal/Desktop
    - Configuration
    - Frameworks
    - iosApp
    - Products
  - Frameworks

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CMPUnitConverter main

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  - local.properties
  - README.md
  - settings.gradle.kts
- External Libraries

shared

- shared
  - build
  - schemas
  - src
    - androidMain
    - commonMain
      - composeResources
        - drawable
        - values
        - values-de-rDE
      - kotlin
        - de
          - thomaskuenneth
            - cmpunitconverter
    - desktopMain
    - iosMain
  - build.gradle.kts
  - rules.pro

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Shared UI & Resources

- Composables live in `commonMain`. One UI implementation for all three platforms
- Icons and strings go in `composeResources`. You reference them with the generated `Res` API
- For example, `Res.string.app_name`, `Res.drawable.ic_thermostat`

```
@Composable
fun About(modifier: Modifier = Modifier) {
    Column(
        modifier = modifier.padding(8.dp),
        verticalArrangement = Arrangement.Center,
        horizontalAlignment = Alignment.CenterHorizontally
    ) {
        Image(
            painter = painterResource(Res.drawable.artwork_no_background), null,
            modifier = Modifier.size(96.dp)
        )
        Text(
            text = stringResource(Res.string.app_name),
            modifier = Modifier.padding(top = 16.dp),
            color = MaterialTheme.colorScheme.onSurface
        )
        Text(
            text = "$VERSION ($BUILD_NUMBER)",
            color = MaterialTheme.colorScheme.onSurface
        )
        Text(
            text = platformName,
            style = MaterialTheme.typography.bodySmall,
            textAlign = TextAlign.Center,
            color = MaterialTheme.colorScheme.onSurface
        )
    }
}
```

ViewModel & Koin

- The converter ViewModels (e.g. `TemperatureViewModel`, `DistanceViewModel`) hold the math and state
- Defined in a *Koin* module in `commonMain`
- The UI gets them via `koinViewModel()`
- They don't know (or care) if they're running on an iPhone or a Pixel

Android

- appleMain
- appleTest
- commonMain
  - kotlin
    - de.thomaskuenneth.cmpunitconverter
      - app
      - composables
      - di
        - AppModule.kt
        - KoinApp.kt
      - distance
        - DistanceRepository.kt
        - DistanceSupportingPaneUseCase
        - DistanceViewModel
      - temperature
      - AbstractBaseRepository
      - AbstractConverterViewModel
      - AbstractSupportingPaneUseCase
      - AppDestinations
      - AppIcons

AppModule.kt

```kotlin
package de.thomaskuenneth.cmpunitconverter.di

import ...

1 Usage
val appModule = module {
    single<AppDatabase> { getRoomDatabase( builder = getDatabaseBuilder()) }
    single<HistoryDao> { get<AppDatabase>().getHistoryDao() }

    singleOf( constructor = ::HistoryRepository)

    singleOf( constructor = ::AppRepository)
    viewModelOf( constructor = ::AppViewModel)

    singleOf( constructor = ::TemperatureRepository)
    singleOf( constructor = ::TemperatureSupportingPaneUseCase)
    viewModelOf( constructor = ::TemperatureViewModel)

    singleOf( constructor = ::DistanceRepository)
    singleOf( constructor = ::DistanceSupportingPaneUseCase)
    viewModelOf( constructor = ::DistanceViewModel)
}
```

Platform Entry Points

- Android: Application calls `initKoin` at startup; `MainActivity` calls `setContent`. iOS: `ComposeView` inside a `ViewController`. Desktop: `main()` launches the window
- One quirk: Kotlin/Native exports `Unit`-returning top-level functions with a `do` prefix in Swift
- So, you have `initKoin()` in Kotlin but call `doInitKoin()` from Swift

```swift
import SwiftUI
import ComposeApp

@main
struct iOSApp: App {

    init() {
        // Kotlin/Native exports Unit-returning top-level functions with a "do" prefix in Swift/Obj-C.
        // Shared code has initKoin(); the generated API is doInitKoin(). See Koin CMP quickstart:
        // https://insert-koin.io/docs/quickstart/cmp
        KoinAppKt.doInitKoin()
    }

    var body: some Scene {
        WindowGroup {
            ContentView()
        }
    }
}
```

Lessons from the trenches

- Keep Compose Multiplatform, Compose Compiler plugin, and Kotlin Multiplatform plugin in sync. The docs have tables—use them
- Your Android app's `activity-compose` must match the Compose version the shared module uses. If they're out of sync, you can get `NoSuchMethodError` at `setContent`, and other runtime surprises
- When you upgrade, check the compatibility page first. It saves a lot of debugging

- When starting a new project, use the AGP 9 structure from day one
- Migrating later works but is quite some work
- Put platform quirks behind `expect`/`actual` and document them (like `doInitKoin`)
- Keeps shared code clean and stops people from fixing what isn't broken

Time to wrap things up: what I showed you

- **The story**: Jetpack history, the Jungle, the golden path, and the twist—same stack, beyond Android
- **The lineup**: *Compose Multiplatform*, *ViewModel*, *Navigation*, *Room*, *DataStore*. All in KMP
- **How it fits**: one shared module, thin platform apps, entry points, DI
- **The case study**: CMP Unit Converter—one codebase, Android, iOS, Desktop. Clone it and try it

What I left out

- **Testing**: you can unit test ViewModels, repositories, and use cases in `commonTest`. Room: in-memory DB, same DAOs. expect/actual: test doubles. The docs and the sample repo have the details
- **Version alignment**: (we had a slide on it) `activity-compose` and Compose versions must match. When you upgrade, check the compatibility tables
- **What's not (yet) in KMP**: WorkManager, CameraX, and some others are still Android-only. For those you keep them in the app module or find alternatives

Jetpack roadmap beyond Android

- Google and JetBrains did a great job in unbundling Jetpack from the Android platform
- More libraries will get KMP support. The golden path stack is already stable and production-ready
- One architecture, many platforms—that's the direction. Investing in this stack isn't a bet on a niche

Thank you!

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https://github.com/tkuenneth/CMP-Unit-Converter

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Feedback? Yes, please

Refuelling your Jetpack - using Jetpack libraries beyond Android

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